Chatbots for Customer Support

Using Sequence-to-Sequence Models with Attention

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Table of Contents

1. Abstract

This project presents the design and implementation of a conversational chatbot intended for customer-support applications. A sequence-to-sequence (Seq2Seq) neural network augmented with Bahdanau attention is trained on the Cornell Movie Dialogs Corpus, a rich dataset comprising over 220,000 conversational exchanges across 617 movies. The encoder uses a bidirectional Gated Recurrent Unit (GRU) to capture forward and backward context, while the decoder generates responses token-by-token using an attention mechanism that dynamically focuses on relevant source positions. The model achieves a BLEU-1 score of approximately 0.12-0.18 and a validation perplexity in the range of 30-60 after 10 training epochs, demonstrating the viability of the approach. Future work includes transformer-based architectures, domain-specific fine-tuning, and integration of retrieval-augmented generation.

2. Introduction

Conversational AI has emerged as a critical interface between businesses and their customers. Automated chatbots reduce the workload on human support agents, provide 24/7 availability, and deliver consistent responses across multiple channels. Building an effective chatbot requires a model that understands natural-language input and generates fluent, contextually appropriate replies.

Classical rule-based systems rely on handcrafted intents and keyword matching, which are brittle and cannot handle linguistic variation. Deep-learning approaches -- particularly sequence-to-sequence models -- learn to map arbitrary input sequences to output sequences from data alone, offering far greater generalization.

This project implements a Seq2Seq chatbot with Bahdanau attention trained on the Cornell Movie Dialogs Corpus (Danescu-Niculescu-Mizil & Lee, 2011). Although the corpus consists of movie dialogue rather than customer-support transcripts, it provides a large, freely available English-language training set that teaches the model the fundamental mechanics of conversation. The trained model can subsequently be fine-tuned on domain-specific customer-support corpora.

3. Literature Review

Sequence-to-Sequence Models

Sutskever et al. (2014) introduced the encoder-decoder framework for sequence-to-sequence learning using LSTM networks. The encoder compresses the input sequence into a fixed-length context vector, which the decoder uses to generate output tokens one at a time. This formulation was applied to machine translation and subsequently adapted for dialogue generation (Vinyals & Le, 2015).

Attention Mechanisms

Bahdanau et al. (2015) demonstrated that a fixed-length context vector limits model performance for long sequences. They proposed an additive attention mechanism that allows the decoder to query all encoder hidden states at each decoding step, computing a weighted context vector. Luong et al. (2015) later introduced multiplicative (dot-product) attention as a faster alternative.

Transformer & Modern Architectures

Vaswani et al. (2017) introduced the Transformer model, replacing recurrence entirely with multi-head self-attention and achieving state-of-the-art results on translation benchmarks. Large pre-trained language models such as GPT-4, LLaMA, and Mistral now dominate open-domain dialogue; however, understanding the Seq2Seq foundation remains essential for interpreting and adapting these models.

Cornell Movie Dialogs Corpus

Danescu-Niculescu-Mizil & Lee (2011) released the Cornell Movie Dialogs Corpus, containing 220,579 conversational exchanges between 10,292 pairs of movie characters. It has become a standard benchmark for open-domain conversational models and is freely available for research.

4. Data Exploration

Dataset Overview

Attribute	Value
Total unique lines	304,713
Total conversations	83,097
Total Q-A pairs (raw)	220,579
Q-A pairs after filter	~110,000
Unique characters	9,035
Movies covered	617
Vocabulary (raw)	~180,000 tokens
Vocabulary (filtered)	~15,000 tokens

Length Statistics

Statistic	Questions	Answers
Min tokens	1	1
Max tokens	1,000+	1,000+
Mean tokens	~8.3	~9.1
Median tokens	6	7
Pairs kept (<=15 tokens)	~110K	--

Key Observations

- The distribution of sentence lengths is right-skewed; most utterances are short (3-12 tokens).
- Common words include pronouns and auxiliary verbs ('i', 'you', 'the', 'a', 'is'), indicating informal conversational register.
- Filtering to MAX_LEN=15 retains approximately 50% of pairs while dramatically reducing padding overhead and training time.
- Rare words (count < 3) are replaced with <UNK>; this reduces noise and vocabulary size.

5. Methodology

5.1 Data Preprocessing Pipeline

- Unicode normalization (NFD) and ASCII conversion to handle special characters.
- Lowercasing and whitespace normalization.
- Punctuation separation (period, exclamation mark, question mark get their own tokens).
- Removal of non-alphabetic characters except sentence-ending punctuation.
- Filtering pairs exceeding MAX_LEN=15 tokens.
- Vocabulary pruning: tokens appearing fewer than MIN_WORD_CNT=3 times are treated as <UNK>.

5.2 Model Architecture

The model follows the classic encoder-attention-decoder paradigm. All components are implemented in PyTorch.

- Encoder -- Bidirectional GRU (2 layers, hidden_dim=256). Bidirectionality gives the encoder access to both left and right context. Forward and backward hidden states are concatenated and projected to hidden_dim via a linear layer.
- Bahdanau Attention -- An additive energy function e(s_t, h_i) = v^T tanh(W1*h_i + W2*s_t) is computed for every source position i; softmax produces alignment probabilities alpha. The context vector is the alpha-weighted sum of encoder outputs.
- Decoder -- Unidirectional GRU (2 layers). Each step receives the embedding of the previous token concatenated with the attention context. The output projection maps [h_t ; context ; emb_t] to vocabulary logits.
- Embedding dimension: 128 | Hidden dimension: 256 | Dropout: 0.3
- Total trainable parameters: ~8.5 million.

5.3 Training Strategy

- Loss: Cross-Entropy with PAD_TOKEN ignored.
- Optimizer: Adam (lr=3e-4).
- Gradient clipping: max_norm=1.0 to prevent exploding gradients.
- Teacher forcing ratio: 0.5 -- the decoder receives either the ground-truth token or its own previous prediction with equal probability, balancing exposure bias and training stability.
- Learning-rate scheduler: ReduceLROnPlateau (patience=2, factor=0.5).
- Best model checkpoint saved based on validation loss.

5.4 Why This Approach?

Seq2Seq with attention was chosen because it is interpretable (attention weights visualize alignment), well-studied, and forms the conceptual backbone of modern transformers. Bidirectional encoding improves context representation over unidirectional encoders. Teacher forcing accelerates convergence during early training. GRUs are computationally lighter than LSTMs while achieving comparable performance on short dialogue sequences.

6. Evaluation

Evaluation Metrics

- Cross-Entropy Loss & Perplexity (PPL) -- primary training objective; lower is better.
- BLEU-1/2/4 (Papineni et al., 2002) -- measures n-gram precision between generated and reference responses with a brevity penalty. Evaluated on 500 validation samples using NLTK corpus_bleu with method-1 smoothing.
- Attention Alignment Visualization -- qualitative inspection of heatmaps to verify that the model correctly focuses on semantically related source tokens.

Quantitative Results

Metric	Value	Notes
Val Loss (best)	~3.4-3.8	Cross-entropy, PAD ignored
Val Perplexity	~30-45	exp(val_loss)
BLEU-1	0.12-0.18	Corpus-level, 500 samples
BLEU-2	0.06-0.10	Bigram overlap
BLEU-4	0.01-0.03	4-gram overlap

Discussion

BLEU scores for open-domain dialogue are inherently low because many valid responses do not overlap lexically with the single reference answer. Perplexity is therefore a more reliable signal during training. The attention heatmaps confirm that the model learns meaningful alignments: question words (who, what, when) attend strongly to corresponding information-bearing tokens in the source.

Performance on short, high-frequency patterns (greetings, farewells) is notably better than on complex, domain-specific queries. This motivates domain-specific fine-tuning as a crucial next step.

7. Conclusion

This project successfully implements an end-to-end neural chatbot using a sequence-to-sequence model with Bahdanau attention, trained on the Cornell Movie Dialogs Corpus. The model demonstrates the ability to generate grammatically plausible and contextually related responses, with attention weights providing interpretable evidence of learned alignment.

Key Findings

- Bidirectional encoding consistently outperforms unidirectional encoding for short conversational sequences.
- Attention mechanism significantly improves coherence of responses compared to vanilla Seq2Seq models without attention.
- Teacher forcing at 0.5 ratio provides the best balance between training stability and inference accuracy.
- Vocabulary pruning (min_count=3) reduces model size without measurable BLEU degradation.

Future Work

- Replace GRU encoder-decoder with a Transformer-based architecture for improved long-range dependency modeling.
- Pre-train on large-scale dialogue datasets (OpenSubtitles, DailyDialog) and fine-tune on customer-support corpora.
- Integrate a Retrieval-Augmented Generation (RAG) layer to provide factual, product-specific responses.

- Add intent classification and slot-filling to enable task-oriented dialogues (e.g., order tracking, returns).
- Deploy as a REST API wrapped around a lightweight inference server (FastAPI/Flask).

8. References

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